

MODULE 6

Areas of Application of Computers

Comprehensive Lecture Series | Academic Year 2024

| COMPUTERS IN SOCIETY

The ubiquity of computers has redefined how modern society functions. We no longer treat computers as "specialized tools" but as **essential utilities**, similar to electricity or water.

Core Impact Areas

- ✓ Communication Velocity
- ✓ Data Processing Scale
- ✓ Accessibility to Knowledge

| CBT & E-LEARNING



CBT Systems

Self-paced learning modules that provide instant feedback and interactive assessment.



Massive Open Online Courses

Platforms like Coursera, Khan Academy, and edX democratizing Ivy League education.

| VIRTUAL CLASSROOMS

- ✓ **LMS Integration:** Canvas and Google Classroom managing assignments and grading.
- ✓ **Synchronous Learning:** Real-time video conferencing via Zoom and Microsoft Teams.
- ✓ **Collaborative Tools:** Shared whiteboards and live document editing for group projects.

| RESEARCH & INFORMATION

Digital Libraries

Access to millions of peer-reviewed journals and historical archives via JSTOR, IEEE, and Project Gutenberg.

Global Databases

Search engines and knowledge graphs providing the sum of human knowledge in milliseconds.

| ADMINISTRATIVE EFFICIENCY

Function	Traditional System	Computerized System
Grading	Manual Calculation	Automated GPA Analysis
Records	Physical Files	Encrypted SQL Databases
Enrollment	Queue-based	Instant Web Portal

| SPECIAL NEEDS EDUCATION

Visual Aids

Screen readers (JAWS) and Braille displays for the visually impaired.

Speech Tech

Speech-to-text and predictive text for individuals with motor impairments.

| THE FUTURE: AI & VR

AI Tutors: Personalized learning paths that adapt difficulty based on student performance in real-time.

VR Field Trips: Immersive historical or biological simulations without leaving the classroom.

| BUSINESS: E-COMMERCE

- ✓ **Global Marketplaces:** Platforms like Amazon and Alibaba connecting buyers and sellers worldwide.
- ✓ **Digital Payments:** Secure transaction gateways (Stripe, PayPal) and blockchain tech.
- ✓ **B2B Logistics:** Automated supply chains reducing delivery lead times.

| MARKETING & BIG DATA

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QUINTILLION BYTES

Data created daily for consumer profiling.

Predictive Analytics

Using machine learning to anticipate customer needs before they even search for a product.

| INVENTORY MANAGEMENT



| MANUFACTURING TECH

Term	Meaning	Application
CAD	Computer Aided Design	Architectural & Product Drafting
CAM	Computer Aided Manufacturing	CNC Milling & 3D Printing

| ROBOTICS IN INDUSTRY

Assembly Lines

Precision welding and sorting at speeds impossible for humans.

Dangerous Environments

Robots handling toxic waste or extreme temperatures safely.

| REMOTE WORK EVOLUTION



Collaboration

Slack and Trello managing projects across time zones.



Security

VPNs and Zero-Trust architecture protecting home offices.

FINANCIAL TECHNOLOGY

ATM Transactions

85%

High-Freq Trading

95%

Cashless Payments

70%

SPECIALIZED FIELDS

Government, Science, and Beyond

| GOVERNMENT APPLICATIONS

- ✓ **E-Governance:** Online renewal of licenses and tax filing (IRS, FIRS).
- ✓ **National Security:** Biometric databases and ID systems.
- ✓ **Census Data:** Massive demographic analysis for policy making.

| SCIENCE: WEATHER MODELING

Supercomputers process millions of data points from satellites to predict storms and track long-term climate change with high accuracy.

Data Sources

Barometric pressure, wind speeds, and ocean temperature gradients.

| SCIENCE: SPACE EXPLORATION



Satellite Control

Maintaining orbit and data transmission for GPS and Comms.



Simulations

Simulating landing conditions on Mars before actual mission launch.

| ENGINEERING: STRUCTURAL

- ✓ **Stress Analysis:** Testing bridge stability in 3D simulations.
- ✓ **BIM:** Building Information Modeling for architectural coordination.
- ✓ **Fluid Dynamics:** Testing aircraft aerodynamics in virtual wind tunnels.

| TRANSPORT: ATC SYSTEMS



Radar
Detection

Processing
Trajectory

Conflict
Avoidance

Guidance
Safe Landing

| GPS & AUTONOMOUS VEHICLES

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SATELLITES

Providing global positioning accuracy within centimeters.

Self-Driving Tech

Real-time processing of LiDAR, Radar, and Cameras to navigate city streets without human input.

| MODERN COMMUNICATIONS

VOIP

Voice over Internet Protocol (WhatsApp, Skype) replacing traditional telephony.

Social Media

Global networking and real-time information dissemination platforms.

| RECREATION: GAMING



Cloud Gaming

Streaming high-end games to low-spec devices via high-speed web.



Immersive Reality

Full-sensory VR experiences and AR (Augmented Reality) gaming.

| CGI AND CINEMATOGRAPHY

- ✓ **Motion Capture:** Translating human movement into digital characters.
- ✓ **Ray Tracing:** Rendering light and shadows realistically in movies.
- ✓ **Post-Processing:** Digital color grading and complex visual effects.

| MILITARY: DRONE TECH

Unmanned Aerial Vehicles (UAVs) provide surveillance and tactical capabilities without risking personnel in combat zones.

Remote Operations

Commanding assets from thousands of miles away using satellite links.

| MILITARY STRATEGY

AES-256

ENCRYPTION STANDARD

The core of modern battlefield communication security.

| HEALTHCARE APPLICATIONS

- ✓ **Telemedicine:** Remote consultations and chronic disease monitoring.
- ✓ **Robotic Surgery:** Da Vinci systems enabling sub-millimeter precision.
- ✓ **EHR:** Digital patient history accessible across different hospitals.

| MEDICAL IMAGING

Modality	Benefit
MRI / CT Scan	3D visualization of internal organs
AI Diagnostics	Detecting cancer signs earlier than humans

| SUMMARY: MODULE 6

Efficiency

Computers maximize output in every sector from farming to medicine.

Innovation

New fields (FinTech, SpaceTech) are born entirely from computing power.